

Sulzer AHLSTAR A22-50 Centrifugal Pump — Maintenance Manual

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EQUIPMENT

Tag: EQ-PMP-A22-50-01 Plant: North Refinery Unit 4
Model: AHLSTAR A22-50 Category: Centrifugal Pump
Serial: AH-2023-44129
Manufacturer: Sulzer (Switzerland)

SAFETY HAZARDS

- Stored mechanical energy
- Hot surfaces (process side)
- Pinch points

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Section 4 — Seal Replacement

Procedure ID: PRC-PMP-001-A

Category: Corrective Maintenance Severity: Medium

Performed by: T-1001 Jordan Reyes (Rotating Equipment)

OVERVIEW

This procedure covers replace mechanical shaft seal on equipment EQ-PMP-A22-50-01 (AHLSTAR A22-50). Follow the safety hazards listed on page 1 before starting. Lock out and tag out upstream isolation before disassembly.

PARTS REQUIRED

- SEAL-MS-25-A — Mechanical Shaft Seal 25mm Type A (Seals)
- GSKT-ANSI-150 — ANSI 150 Spiral-Wound Gasket DN50 (Gaskets)
- BLT-M16-HEX — Hex Bolt M16 x 60 Grade 8.8 (Fasteners)

STEPS

1. Confirm isolation per plant LOTO procedure. Verify zero energy state.
2. Stage the parts listed above plus a calibrated torque wrench and clean rags.
3. Disassemble per Section 4 of this manual.
4. Inspect mating surfaces for wear, scoring, or corrosion. Photograph any anomalies.
5. Install replacement parts in the order listed. Torque fasteners per Table 2.
6. Reassemble and reinstate auxiliaries (lubrication, cooling, instrumentation).
7. Remove isolation and perform a no-load function test.
8. Record completion against work order in CMMS.

ACCEPTANCE CRITERIA

- No abnormal vibration or noise during no-load run.
- All fasteners torqued and witness-marked.
- No leaks at static or dynamic seals after a 5-minute run.

Section 5 — Lubrication Schedule

Procedure ID: PRC-PMP-001-B

Category: Preventive Maintenance Severity: Low

Performed by: T-1001 Jordan Reyes (Rotating Equipment)

OVERVIEW

This procedure covers quarterly bearing lubrication on equipment EQ-PMP-A22-50-01 (AHLSTAR A22-50). Follow the safety hazards listed on page 1 before starting. Lock out and tag out upstream isolation before disassembly.

PARTS REQUIRED

- BRG-6206-2RS — Sealed Deep-Groove Ball Bearing 30mm (Bearings)
- GRSE-LITH-EP2 — Lithium EP2 Grease 400g Cartridge (Lubricants)

STEPS

1. Confirm isolation per plant LOTO procedure. Verify zero energy state.
2. Stage the parts listed above plus a calibrated torque wrench and clean rags.
3. Disassemble per Section 5 of this manual.
4. Inspect mating surfaces for wear, scoring, or corrosion. Photograph any anomalies.
5. Install replacement parts in the order listed. Torque fasteners per Table 2.
6. Reassemble and reinstate auxiliaries (lubrication, cooling, instrumentation).
7. Remove isolation and perform a no-load function test.
8. Record completion against work order in CMMS.

ACCEPTANCE CRITERIA

- No abnormal vibration or noise during no-load run.
- All fasteners torqued and witness-marked.
- No leaks at static or dynamic seals after a 5-minute run.

Section 6 — Impeller Inspection

Procedure ID: PRC-PMP-001-C

Category: Reliability Inspection Severity: Medium

Performed by: T-1208 Aisha Kone (Reliability Engineering)

OVERVIEW

This procedure covers impeller balance and run-out check on equipment EQ-PMP-A22-50-01 (AHLSTAR A22-50). Follow the safety hazards listed on page 1 before starting. Lock out and tag out upstream isolation before disassembly.

PARTS REQUIRED

- IMPLR-CAST-200 — Cast Stainless Impeller 200mm (Wet End)

STEPS

1. Confirm isolation per plant LOTO procedure. Verify zero energy state.
2. Stage the parts listed above plus a calibrated torque wrench and clean rags.
3. Disassemble per Section 6 of this manual.
4. Inspect mating surfaces for wear, scoring, or corrosion. Photograph any anomalies.
5. Install replacement parts in the order listed. Torque fasteners per Table 2.
6. Reassemble and reinstate auxiliaries (lubrication, cooling, instrumentation).
7. Remove isolation and perform a no-load function test.
8. Record completion against work order in CMMS.

ACCEPTANCE CRITERIA

- No abnormal vibration or noise during no-load run.
- All fasteners torqued and witness-marked.
- No leaks at static or dynamic seals after a 5-minute run.